

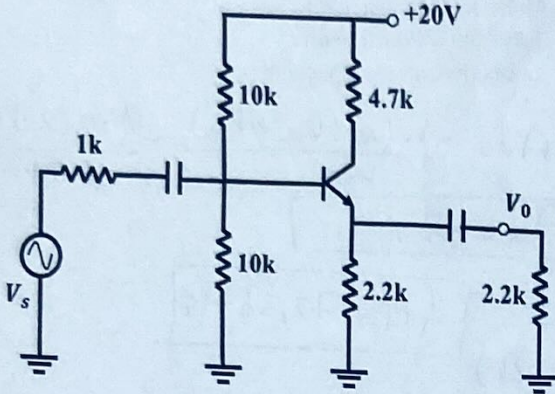
Adı Soyadı: **CEVAP**
Numarası: **ANAHTARI**

Konya Teknik Üniv., Müh. ve Doğa Bil. Fak.,
Elk.-Elt. Müh. Böl., Elektronik-II VİZE Sınavı

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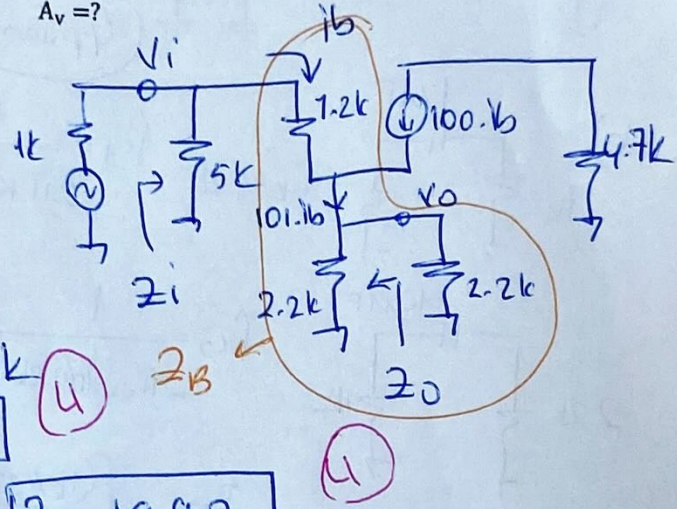
Tarih: 05.04.2024 Süre : 90 dak. BAŞARILAR... Prof. Dr. Seral ÖZŞEN & Prof. Dr. Murat CEYLAN

S.1.



$$h_{fe} = 100, h_{ie} = 1.2k$$

- a) $Z_i, Z_o, A_v, A_{v_s}, A_i = ?$
b) $T_1 = T_2$ olacak şekilde darlington bağlanırsa $A_v = ?$



$$a) V_i = 1.2k i_b + 101 \cdot i_b (1.1k)$$

$$V_i = 112,3k i_b$$

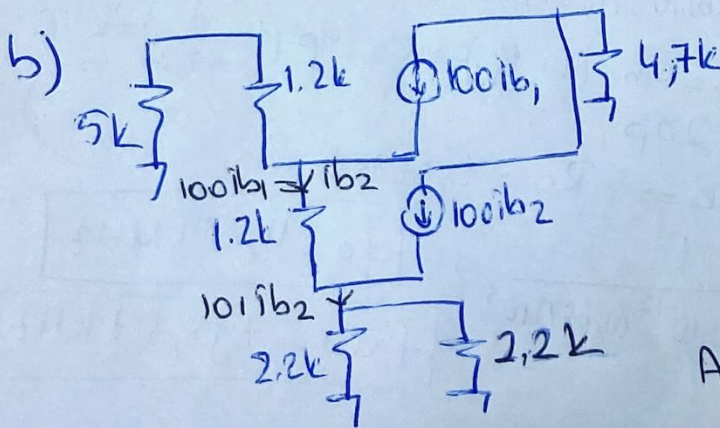
$$Z_B = \frac{V_i}{i_b} = 112,3k \quad Z_i = 5k // 112,3k \quad Z_i = 4,78k$$

$$Z_o = \left[\frac{(1k // 5k) + 1.2k}{101} \right] // 2.2k \Rightarrow Z_o = 19,9\Omega$$

$$A_v = \frac{V_o}{V_i} = \frac{101 \cdot i_b (1.1k)}{112,3k i_b} \Rightarrow A_v = 0,989$$

$$A_i = A_v \cdot \frac{Z_i}{R_L} = 0,989 \cdot \frac{4,78k}{2.2k} \quad A_i = 2,148$$

$$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s} = 0,989 \cdot \frac{4,78k}{5,78k} \quad A_{v_s} = 0,817$$



$$V_i = 1,2k i_{b1} + 1,2k \cdot 101 i_{b1} + (1.1k) 100 i_{b2}$$

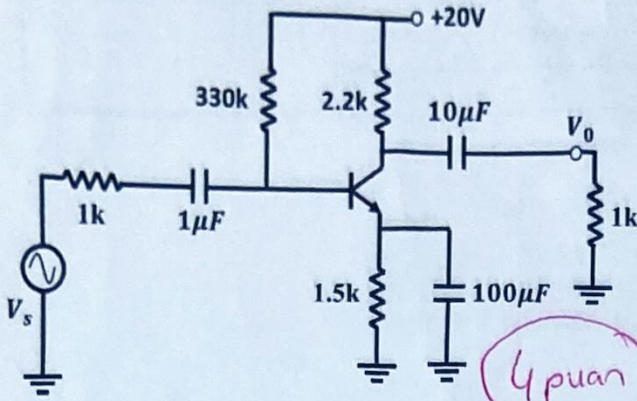
$$i_{b2} = 101 i_{b1} \text{ ise,}$$

$$V_i = 11343,5k i_{b1}$$

$$V_o = 101 i_{b2} (1.1k)$$

$$A_v = \frac{V_o}{V_i} = \frac{101 \cdot 101 i_{b1} (1.1k)}{11343,5k i_{b1}} \quad A_v = 0,989$$

5 puan



$$h_{ie} = 1.2k, h_{fe} = 150$$

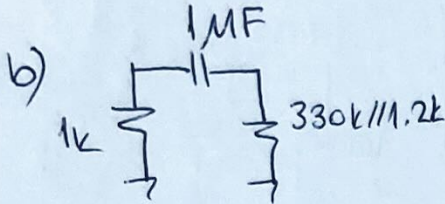
$$C_{bc} = 4pF, C_{be} = 8pF, C_{ce} = 6pF$$

$$C_{w1} = C_{w2} = 10pF$$

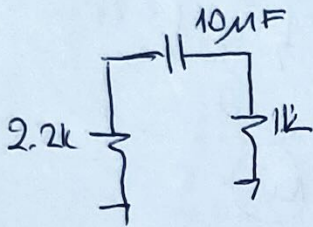
- a) Orta frekans bölgesinde $A_v = ?$
 b) Alt kesim frekansı nedir?
 c) Üst kesim frekansı nedir?
 d) Kazanç-frekans eğrisini çiziniz.

$$a) A_v = \frac{-h_{fe}(R_c \parallel R_y)}{h_{ie}} = \frac{-150(2.2k \parallel 1k)}{1.2k}$$

$$A_v = -85.94$$

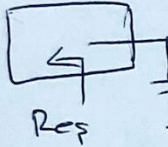


$$f_{c1} = \frac{1}{2\pi \cdot 10^{-6} \cdot (1k + 1.2k)} \quad \boxed{f_{c1} = 72.38 \text{ Hz}}$$



$$f_{c2} = \frac{1}{2\pi \cdot 10^{-6} \cdot (2.2k + 1k)} \quad \boxed{f_{c2} = 4.97 \text{ Hz}}$$

9 puan



$$R_{eq} = \left[\frac{(1k \parallel 330k) + 1.2k}{151} \right] \parallel 1.5k = 14.4\Omega$$

$$f_{c3} = \frac{1}{2\pi \cdot 100 \cdot 10^{-6} \cdot (14.4\Omega)} \Rightarrow \boxed{f_{c3} = 110.58 \text{ Hz}}$$

$$\boxed{f_{alt} = 110.58 \text{ Hz}}$$

c) $C_i = C_{be} + C_{w1} + C_1$ $C_1 = C_{bc}(1 - A_v) \Rightarrow C_1 = 4p(1 - (-85.94)) = 347.76p$

$$C_i = 8p + 10p + 347.76p \Rightarrow C_i = 365.76p$$

$$R_i = R_s \parallel R_B \parallel h_{ie} = 1k \parallel 330k \parallel 1.2k \Rightarrow R_i \approx 0.545k$$

$$f_i = \frac{1}{2\pi \cdot C_i \cdot R_i} = \frac{1}{2\pi \cdot (365.76) \cdot 10^{-12} \cdot (0.545) \cdot 10^3} \Rightarrow \boxed{f_i = 798.817 \text{ kHz}}$$

$$C_o = C_{ce} + C_{w2} + C_2$$

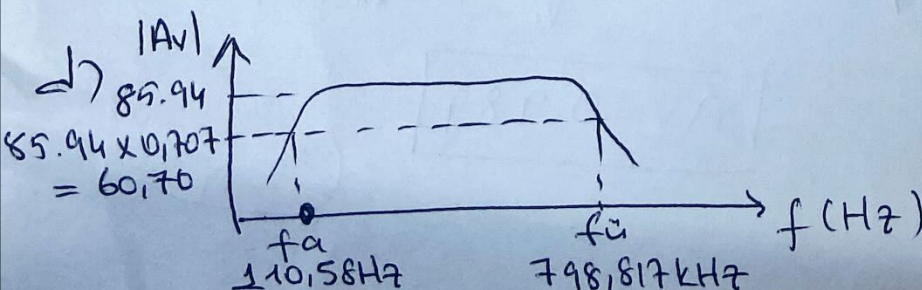
$$C_o = 6p + 10p + 4p \Rightarrow C_o = 20p$$

$$R_o = R_c \parallel R_y = 2.2k \parallel 1k \Rightarrow R_o = 0.6875k$$

9 puan

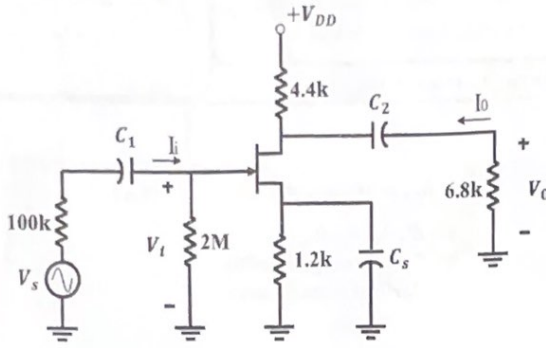
$$f_o = \frac{1}{2\pi \cdot C_o \cdot R_o} = \frac{1}{2\pi \cdot 20 \cdot 10^{-12} \cdot (0.687) \cdot 10^3} \Rightarrow \boxed{f_o = 11.589 \text{ MHz}}$$

$$\boxed{f_{üst} = 798.817 \text{ kHz}}$$



3 puan

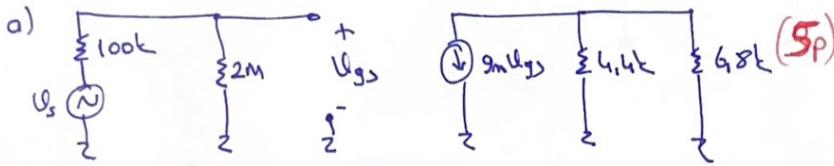
S.2.



Yandaki devrede $g_m = 3.8mS$ verilmiştir. Buna göre;

- Devrenin AC eşdeğerini çiziniz.
- $Z_i = ?$ $Z_o = ?$
- $A_v = \frac{V_o}{V_i} = ?$ $A_{v_s} = \frac{V_o}{V_s} = ?$
- $A_i = \frac{I_o}{I_i} = ?$

Gözüm:



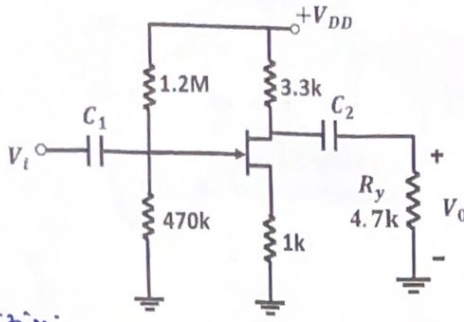
b) $Z_i = 2M\Omega$ (4p) $Z_o = 4.4k\Omega$ (4p)

c) $A_v = \frac{V_o}{V_i} = \frac{-g_m V_{gs} (4.4k\Omega \parallel 6.8k\Omega)}{V_{gs}} \Rightarrow A_v = -10.15$ (4p)

$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s} = -10.15 \cdot \frac{2000k}{2100k} \Rightarrow A_{v_s} = -9.67$ (4p)

d) $A_i = -A_v \cdot \frac{Z_i}{R_L} = 10.15 \cdot \frac{2000k}{6.8k} \Rightarrow A_i \approx 2985.29$ (4p)

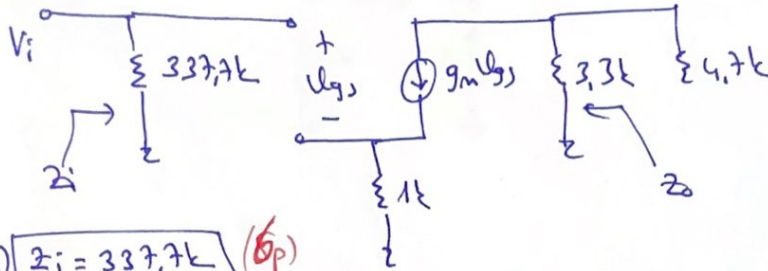
S.3.



Yandaki devrede $g_m = 2.2mS$ verilmiştir. Buna göre;

- $Z_i = ?$ $Z_o = ?$
- $A_v = \frac{V_o}{V_i} = ?$
- $r_d = 50k$ olarak verilirse $Z_i = ?$, $Z_o = ?$, $A_v = ?$

Gözüm:



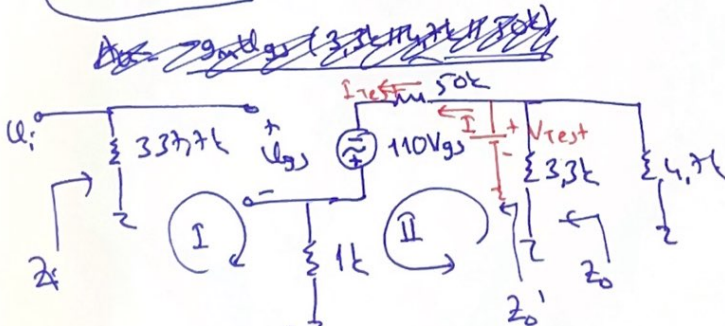
a) $Z_i = 337.7k$ (6p)

$Z_o = 3.3k$ (6p)

b) $A_v = \frac{V_o}{V_i} = \frac{-g_m V_{gs} (3.3k // 4.7k)}{V_{gs} + g_m V_{gs} \cdot 1k} \Rightarrow A_v \approx -1.33$ (6p)

c) $r_d = 50k$ olursa:

$Z_i = 337.7k$ (3p) ~~$Z_i = 3.3k // 50k$~~ ~~$Z_o = 3.3k$~~



$V_o = -I \cdot (3.3k // 4.7k)$

$V_i = V_{gs} + 1k \cdot I$

(II) noluceninden:

$(4.7k // 3.3k) \cdot I + 50k \cdot I - 110V_{gs} + 1k \cdot I = 0$
 $52.94k \cdot I = 110V_{gs}$

$I \approx 2.08mV_{gs}$

$A_v = \frac{-2.08mV_{gs} \cdot 1.94k}{V_{gs} + 2.08mV_{gs} \cdot 1k} \Rightarrow -1.31$

$A_v = -1.31$ (2p)

$Z_o = Z_o' // 3.3k$

$Z_o' = V_{test} / I_{test}$

$V_{test} = 50k \cdot I_{test} - 110V_{gs} + 1k \cdot I_{test}$
 (I) noluceninden ($V_i = 0V$, I_{test})

$V_{gs} + 1k \cdot I_{test} = 0$

$V_{gs} = -1k \cdot I_{test}$

$V_{test} = 51k \cdot I_{test} + 110 \cdot 1k \cdot I_{test}$

$V_{test} = 161k \cdot I_{test}$

$Z_o' = 161k$

$Z_o = 3.3k // 161k$

$Z_o = 3.23k\Omega$ (2p)